Yueh-Po Peng

Experience

Senior Machine Learning Engineer **Jul 2025 – Present**
Gamania Digital Entertainment **Taipei, Taiwan**
AI Manga Creation Platform

- Built a conversational AI agent that collaborates with creators through multi-turn dialogue to develop storylines, then orchestrates sequential generation of characters, locations, and storyboards.
- Designed a retrieval-augmented storyboard system that fetches relevant character/location assets from an indexed library to ensure visual consistency across panels.
- Developed full-stack platform (Python backend + frontend) serving 100+ beta users, reducing manga chapter prototyping from days to under 30 minutes.

AI Vtuber & Voice Cloning

- Trained voice cloning models enabling AI virtual characters to cover songs in target speaker voices with ~30 minutes of training data.
- Built internal tooling for non-technical staff to manage voice model training and deployment workflows.
- Integrated Live2D avatar animation with LLM-driven persona dialogue and voice cloning into a unified interactive system.

Visiting Researcher **Jun 2025 – Oct 2025**
Sony Computer Science Laboratories (Sony CSL) **Tokyo, Japan (Hybrid)**
Research Topics: Music Information Retrieval, Automatic Music Transcription

- Proposed VioPTT, a lightweight cascade model for violin technique-aware transcription with a novel synthetic dataset (MOSA-VPT). Published at ICASSP 2026 **[1]**.
- Investigated transfer learning for violin AMT, showing training from scratch on instrument-specific data can match fine-tuned piano-pretrained models. Published at ISMIR 2025 Late Breaking/Demo **[2]**.

AI Engineer **Oct 2024 – May 2025**
Gate.io **Taipei, Taiwan (Remote)**

- Developed a Text-to-SQL AI agent enabling non-technical teams to access internal data, boosting query efficiency by 20%.
- Developed a fund flows anomaly detection system with LLMs and tree-based models, enhancing financial security.

Research Assistant **Mar 2022 – Oct 2024**
Institute of Information Science, Academia Sinica | MCTLAB | Supervisor: Dr. Li Su **Taipei, Taiwan**
Research Topics: Self-Supervised Learning, Medical Imaging

- Proposed a Transformer-based self-supervised learning method for decoding brain signals (fMRI), achieving an 77% reduction in memory footprint.
- Conducted distributed training experiments on high-resolution 4D medical images (fMRI) using TWCC HPC.
- Proposed a whole-brain feature selection method for decoding musical pitch from fMRI **[5]**.

AI Engineer Intern **Mar 2023 – Jul 2024**
Tomofun - World's leading pet technology company **Taipei, Taiwan**
Research Topics: Computer Vision, Large Language Models, Multimodal Learning

- Developed an automatic short music video generation system for daily pet clips.
- Fine-tuned visual language models (e.g., BLIP), achieving a 20.6% improvement in visual question answering.
- Enhanced LLaVA image inference speed by 250% with only a 3% accuracy reduction.
- Developed APIs for visual language models using llama.cpp/ollama for image-caption pair datasets.

Education

National Taiwan University **Feb 2023 – Jun 2024**
Taipei, Taiwan

- M.S. in Data Science | GPA: 4.09/4.30
- Thesis topic: Whole-Brain Feature Selection Methods for Decoding from fMRI Data

National Taiwan University **Sep 2019 – Jan 2022**

- B.S. in Computer Science and Information Engineering (CSIE) **Taipei, Taiwan**

Research & Projects

Guitar Effect Removal **Collaboration with Positive Grid ML Team**

- Proposed a two-stage method to remove distortion effects from guitar recordings using Positive Grid VST plugins.
- Achieved 20% higher audio quality than the best baseline, rated by 26 professional guitarists.
- Published in DAFx 2024 [4].

Whole Brain fMRI Feature Selection

- Proposed a two-stage method to extract fMRI features and predict musical pitch.
- Demonstrated 2-fold improvement over ROI-based feature selection in fMRI-music analysis.
- Published in ICASSP 2023 [5].

Publications

[1] Wang, T. K.*, **Peng, Y. P.***, Su, L., & Cheung, V. K. M. "VioPTT: Violin Technique-Aware Transcription from Synthetic Data Augmentation," Proc. IEEE Int. Conf. on Acoustics, Speech, and Signal Processing 2026 (ICASSP'26). **(* equally contributed)** Paper
[2] **Peng, Y. P.***, Wang, T. K.*, Su, L., & Cheung, V. K. M. "Is Transfer Learning Necessary for Violin Transcription?," Int. Society for Music Information Retrieval Conf. 2025 (ISMIR'25) - Late Breaking/Demo. **(* equally contributed)** Paper
[3] **Peng, Y. P.**, Cheung, V. K. M., & Su, L. "Whole-Brain Transferable Representations from Large-Scale fMRI Data Improve Task-Evoked Brain Activity Decoding," arXiv preprint arXiv:2507.22378, 2025. Paper
[4] Lee, Y. S.*, **Peng, Y. P.***, Wu, J. T., Cheng, M., Su, L., & Yang, Y. H. "Distortion Recovery: A Two-Stage Method for Guitar Effect Removal," Proc. Int. Conf. Digital Audio Effects 2024 (DAFx'24). **(* equally contributed)** Paper | Demo
[5] Cheung, V. K.*, **Peng, Y. P.***, Lin, J. H., & Su, L. "Decoding Musical Pitch from Human Brain Activity with Automatic Voxel-Wise Whole-Brain FMRI Feature Selection," Proc. IEEE Int. Conf. on Acoustics, Speech, and Signal Processing 2023 (ICASSP'23). **(* equally contributed)** Paper

Skills

- **Languages/Frameworks**: Python, PyTorch, TensorFlow, LangChain, LlamaIndex, Pandas, Scikit-learn, Go, HTML, JavaScript, React, Docker, Slurm, C++, C, Linux.
- **Cloud/Infra**: AWS, GCP, TWCC HPC, Docker, Git, CI/CD.
- **Skillset**: LLM Agents, RAG, Prompt Engineering, Text-to-SQL, Voice Cloning, Self-Supervised Learning, Computer Vision, Music Information Retrieval, Multimodal Learning, Distributed Training.